



Al Bello/Getty Images

Tennis

● ● ● ● ● ● ● ●
Jaime Fernandez-Fernandez

Performance Demands of Tennis

Tennis requires a complex interaction of technical, tactical, and psychological components, as well as several physical (i.e., strength, agility) and energetic demands (i.e., aerobic and anaerobic). Training is instrumental in achieving these requirements. In order to strive toward these needs, coaches and athletes must navigate a journey to optimize the various training inputs (i.e., technical, tactical, psychological) and physical conditioning content, in accordance with the most important limiting performance factors, alongside individual player needs (strengths and weaknesses).

The importance of the technical and tactical requirements for tennis performance place a key limitation on the planning of weekly physical training loads, and such needs must be placed above all others within a hierarchical model of tennis performance. Weekly training volume must be adjusted correspondingly with respective physical conditioning volume remaining comparably low. Therefore, at the elite performance level, it becomes critical for physical training to be combined with the on-court technical and tactical training, while off-court sessions should be used to address individual requirements (13).

Sport Description, Including Factors of Winning

Tennis is a game enjoyed by millions around the world who play recreationally and competitively, coupled with extremely large viewing audiences (i.e., the Australian Open 2017 men's final attracted 4.4 million viewers in Australia and 11 million across Europe). Professional tennis players travel and compete extensively year round, with more than 500 annual international tournaments available on the professional men's and women's calendars, including team events for junior, senior, and wheelchair players (34). At a basic level, tennis requires varying sequences of points, games, and sets to be won in order to win the match. Winning enough points wins a game, winning enough games wins a set, and winning enough sets wins the match. Tennis is a unique sport in that a player can be victorious in a match while winning fewer points than his opponent, who is committing more mistakes or errors.

From a physical standpoint, tennis match play is characterized by intermittent exercise, in which alternate short bouts (4 to 10 s) of high-intensity exercise and short bouts (10 to 20 s) of recovery are interrupted by several resting periods of longer duration (60 to 90 s) (33, 44). The duration of a tennis event often is greater than an hour and, in some cases, lasts for more than 5 h (e.g., Australian Open

2012 men's final was 5 h, 53 min). The typical average match time for a best of three match is 1.5 h (33), and effective playing time (percentage of the total time playing the game) amounts to approximately 20% to 30% (21, 36).

Relative Contribution of Physical Performance

The modern game of tennis evolved from a primarily technical sport in which sport-specific technical skills (e.g., racket and ball handling skills, as well as stroke skills such as serving) were the predominant winning factors to more of a dynamic and explosive sport characterized by higher stroke and serve velocities, requiring notably higher physical demands. Increasing evidence suggests that motor skills such as muscle power, strength, agility, speed, mental strength, and a highly developed neuromuscular coordinating ability correlate with tennis performance (26, 46). Moreover, if a player is not in good condition, essential characteristics in tennis, including technique, coordination, concentration, and tactics become exposed in long matches as premature fatigue impairs virtually all tennis-specific skills (27) (figure 18.1).

Targets of Physical Performance in Tennis

As previously mentioned, physical exertion during tennis involves high-intensity efforts interspersed with periods of variable duration and low-intensity activity (21). Thus, from a physiological point of view, demands alternate between energy provision for bouts of high-intensity work and replenishing energy sources and restoring homeostasis during the recovery intervals (21). The most important actions during tennis match play involve serves and strokes, accelerations, decelerations, and changes of direction, which are likely to be dependent on anaerobic energy pathways, supported largely by an aerobic metabolic background for recovery (36).

From a physiological point of view, the sport of tennis has been well defined using various portable analyzers (i.e., oxygen uptake ($\dot{V}O_2$); blood lactate ([La]); heart rate (HR)) that have described the responses during both real and simulated matches (2, 5, 45). Accordingly, mean HR during competitive matches has been shown to range from between 70% and 80% of maximum (HR_{max}), with peak values around 90% to 100% of HR_{max} . Average $\dot{V}O_2$ values correspond to approximately 50% to



Figure 18.1 The position of the tennis player on the three axes illustrates the relative importance of the three main physical capacities for elite participation. Outside of critical psychological aspects, which are difficult to quantify, the pie chart shows the general relative importance of skills (45%), tactical awareness (20%), and physical capacities (35%) for tennis success.

Adapted from G.A. Nader, "Concurrent Strength and Endurance Training: From Molecules to Man," *Medicine & Science in Sports & Exercise* 38, no. 11 (2006): 1965-1970.

60% of $\dot{V}O_2\text{max}$, with values above 80% of $\dot{V}O_2\text{max}$ recorded during intensive rallies (29). Rating of perceived exertion (RPE) has been reported as ranging from 5 to 7 arbitrary units on a scale of 1 to 10 (9, 28) and 10 to 16 arbitrary units on the Borg 20 point scale (12). Regarding [La], it has been reported to fluctuate throughout matches, increasing acutely in response to long rallies and conditions of limited recovery time, indicative of likely greater reliance on anaerobic glycolysis (12, 36). In cases where [La] has been measured during matches, samples were collected during the change of ends (14, 36), with match durations not reported, leaving results to be highly variable and inconclusive. Regardless, [La] values typically remain low during matches, with mean values reported as ranging from between 1.7 and 3.8 mmol/L (12, 33), despite some elite players reaching more than 8 mmol/L (12). With competition points lasting less than 10 s, and the inclusion of frequent rest periods, the physiological demands of match play are unlikely to lead to large accumulations of [La] where it might relate to acute fatigue during tennis matches (33).

The ability to maintain high technical efficiency during phases of high-intensity intermittent exercise is an important feature of successful tennis player performance. From an early age, tennis players spend many hours mastering their individual sport-specific skills, with technical and tactical training volumes often exceeding 15 to 20 h/wk. Moreover, although successful tennis players may not be as tall as their successful basketball or handball player counterparts, the current average height of the Women's Tennis Association (WTA) top-20 players is 176 cm (5'8"), while the average height for players in the Association of Tennis Professionals (ATP) top 50 is now 185 cm (6'1"). Thus, together with the high amount of on-court training volume, my personal belief is that it is necessary to reduce the supplementary-training volume related to running exercise (e.g., traditional interval running), as supported by the relationship shown between high training loads (i.e., on- and off-court) and several joint injuries (e.g., hip) (24, 37).

Key Weapons, Manipulations, and Surveillance Tools

In this section, the different high-intensity interval training (HIIT) weapons, their manipulations, along with ways by which we can monitor their effective-

ness (surveillance) are presented. Although tennis is a very popular sport, information concerning the effectiveness of training interventions is very scarce, and particularly absent is the impact of HIIT on tennis performance-related variables. Therefore, most of the information presented herein is based on my practical experience acquired through years in the sport and on research shown from other intermittent sports.

HIIT Weapons

As presented throughout the book, there are five different metabolic and/or neuromuscular load targets that can be applied throughout a tennis player's season (figure 1.5) by use of the five HIIT weapons (5). For tennis (figure 18.2), the large majority of these include game-based HIIT (40%, mostly in-season), followed by short intervals (30%, some preseason, but mostly in-season), repeated-sprint training (RST) (20%, in-season only), sprint interval training (SIT) (5%, in-season only), and run-based long intervals (5%, in-season only). With the exception of the run-based short and long intervals, tennis movements are mimicked during all HIIT weapon types, with greater uncertainty of play involved with game-based HIIT (figure 5.31). Of course, this breakdown is a general approach only, which should be modified based on individual player needs.

Manipulations of Interval Training Variables

The main programming strategy I implement at the beginning of a player's season is to routinely plan an appropriate volume of all HIIT weapons and target types throughout the year. Intensity of run-based HIIT is calibrated using the 30-15 IFT (chapter 2), which is conducted at the beginning of the preseason, normally after a few days of familiarization with the training routine. I usually don't change the running intensity too much during the season, which, depending on the format and needs of the player, tends to vary from 85% to 100% of V_{IFT} . Moreover, I try to vary training content as much as possible during the week, combining different HIIT formats. For the preseason of an elite junior player, for example, a possible programming combination might be as follows:

Monday, tennis-based HIIT

Wednesday, run-based short HIIT

Friday, tennis-based HIIT.

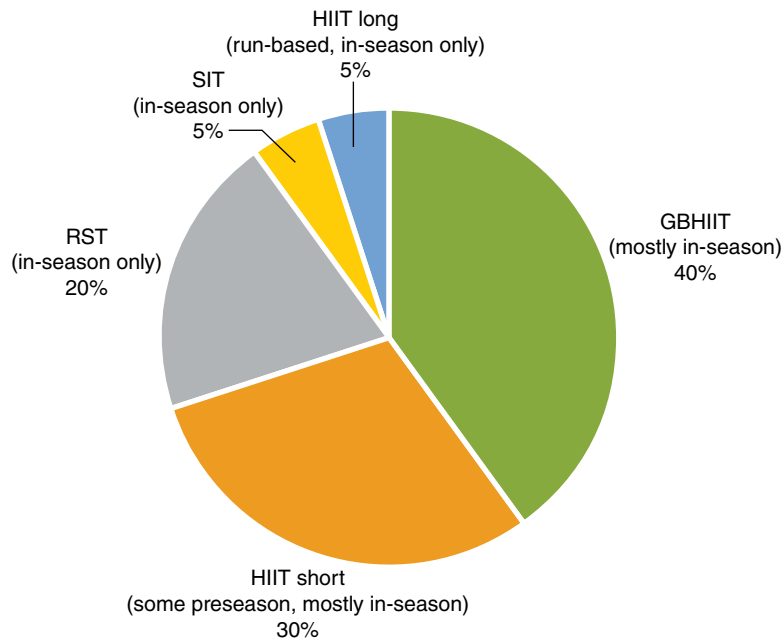


Figure 18.2 Percentage of the different HIIT weapons (formats) used throughout the annual season in elite tennis. Game-based HIIT (40%, mostly in-season), followed by short intervals (30%, some pre-season, but mostly in-season), repeated-sprint training or RST (20%, in-season only), sprint interval training or SIT (5%, in-season only), and run-based long intervals (5%, in-season only). HIIT: high-intensity interval training; RST: repeated-sprint training; SIT: sprint interval training; GBHIIT: game-based HIIT.

HIIT With Long Intervals

Although I don't usually implement run-based HIIT with long intervals in tennis players, there are some exceptions. When a player has an upper-body problem (i.e., injury or overload) or low fitness, I normally implement run-based HIIT with long intervals in the outdoor environment, either on a grass pitch or in the forest (with distances marked using cones), to minimize musculoskeletal strain (5) (HIIT types 3 and 4). Generally, these sessions are performed over 800 to 1000 m efforts completed between 3 and 4 min, at intensities based on the V_{IFT} (80% to 85%), with 2 to 3 min rest, depending on the player's fitness. Normally players are grouped based on their V_{IFT} in order to complete the training sessions in individualized small groups, and typical training volumes range from 3 to 5 repetitions. Generally, these sessions are prescribed at the end of the week before a day off, so players can recover from an anticipated neuromuscular fatigue.

An example of a tennis-specific HIIT with long-interval session is shown in figure 18.3 (17, 40, 43). These involve specific groundstroke/open play from the baseline, with repeated strokes from dif-

ferent positions under pressure, but most specifically during baseline play. An experienced professional coach hand or racket feeds balls for long-HIIT drills. Irrespective of the drill, all players are instructed to move and hit with maximal effort, directing all shots to the target areas near the baseline. These tennis-specific drills differ from game-based drills (see below) due to the absence of an opponent, and in turn, the different decision-making clues and lower certainty involved with the movement patterns required (figure 18.3). Again, intensity is monitored using HR, with target zones set at 90% to 95% of HR_{max} . While overall training volume depends on the player's level, the general aim is to accumulate a minimum of 10 to 15 min of work at intensities close to $\dot{V}O_{2max}$, or in this case, near HR_{max} (5). It is important to acknowledge that although HR monitoring is currently the only (low-cost) method of monitoring HIIT sessions, its effectiveness is limited, as described in chapters 2, 8, and 9. It's additionally important to highlight that using HR to monitor and potentially adjust the intensity of an HIIT session may be limited, as HR cannot inform on the intensity of physical work

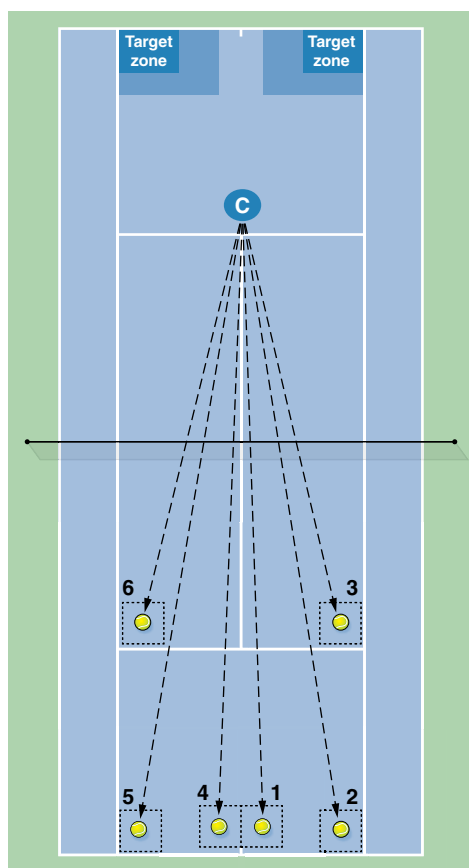


Figure 18.3 Example of tennis-specific HIIT with long intervals (1-2 min work with 45 s to 1 min 30 s rest). P: player; C: coach; 1, 2, 3: forehand strokes; 4, 5, 6: backhand strokes.

Adapted by permission from J. Fernandez-Fernandez, D. Sanz-Rivas, C. Sanchez-Munoz, J.G. de la Aleja Tellez, M. Buchheit, and A. Mendez-Villanueva, "Physiological Responses to On-Court vs Running Interval Training in Competitive Tennis Players," *Journal of Sports Science & Medicine* 10 no. 3 (2011): 540-545.

performed above the speed or power associated with $\dot{V}O_{2\max}$ (5).

HIIT With Short Intervals

My preferred HIIT weapon is the short interval, including 30 s on/30 s off, 20 s/20 s, 15 s/15 s, and 10 s/20 s (see tables 18.1 to 18.4), which target HIIT types 1, 2, 3, and 4 with exercise modality being mostly tennis-specific exercises and run-based HIIT. I have found these formats highly useful as we can fit almost all tennis drills conducted by coaches into practice sessions that integrate the conditioning goal with tennis-based skill development (42). Examples of such drills are shown in figure 18.4.

A possible training session for an elite tennis player might involve 3 sets of 10 repetitions of 30 s work

and 30 s rest, with 2 min rest between sets, alternating between exercises A and B, or 4 sets of 15 repetitions of 15 s work and 15 s rest, with 90 s rest between sets of exercises C and D (see figure 18.4). Normally drills are designed to be performed in several courts, depending on the number of players training at the same time. For example, if we're working with 12 players, three tennis courts can be used, with four players on each court. Players finish the training session in about 40 min.

I usually combine these formats with run-based HIIT, as we can group players in the same facility (normally a 30 m court) based on their V_{IFT} . Thus, we organize running groups and request they run over group-based distances using cones on the court. This can be done easily using the spreadsheet developed by Martin Buchheit (<https://martin-buchheit.net/>) and available from the author on request. These formats are interesting due to their limited level of neuromuscular fatigue and anaerobic contribution (especially 10 s/20 s, HIIT type 2; chapter 5), which is important when players are conducting a specific training session on the same day (or session), as well as when neuromuscular-based sessions (e.g., strength) the following day are planned (5, 7).

Combinations of HIIT formats are also possible, depending on the number of players involved. For example, a group of players might perform on-court tennis drills, while another group performs sets of run-based HIIT adjacent to those players on the lateral sides of the court.

Repeated-Sprint Training

Since RST might be a time-efficient training method of enhancing aerobic adaptations relative to the lower training volume required with other HIIT formats (25), it may represent a useful training strategy for tennis players, especially during the in-season period, when busy schedules limit the number of training sessions available for fitness development. In this regard, I use the RST format (type 4 target) to implement generally 2 to 4 sets of 5 to 8 sprints across a maximum of a 20 m distance, interspersed with 14 to 25 s of passive or active (jog or sport-specific movements, e.g., side steps, crossover steps, etc.) recovery during the first part of the training session. I choose a maximum 20 m distance because relevant speed in tennis involves the ability to move at high velocity over short distances in different directions around the court (33). Indeed, initial acceleration is the key

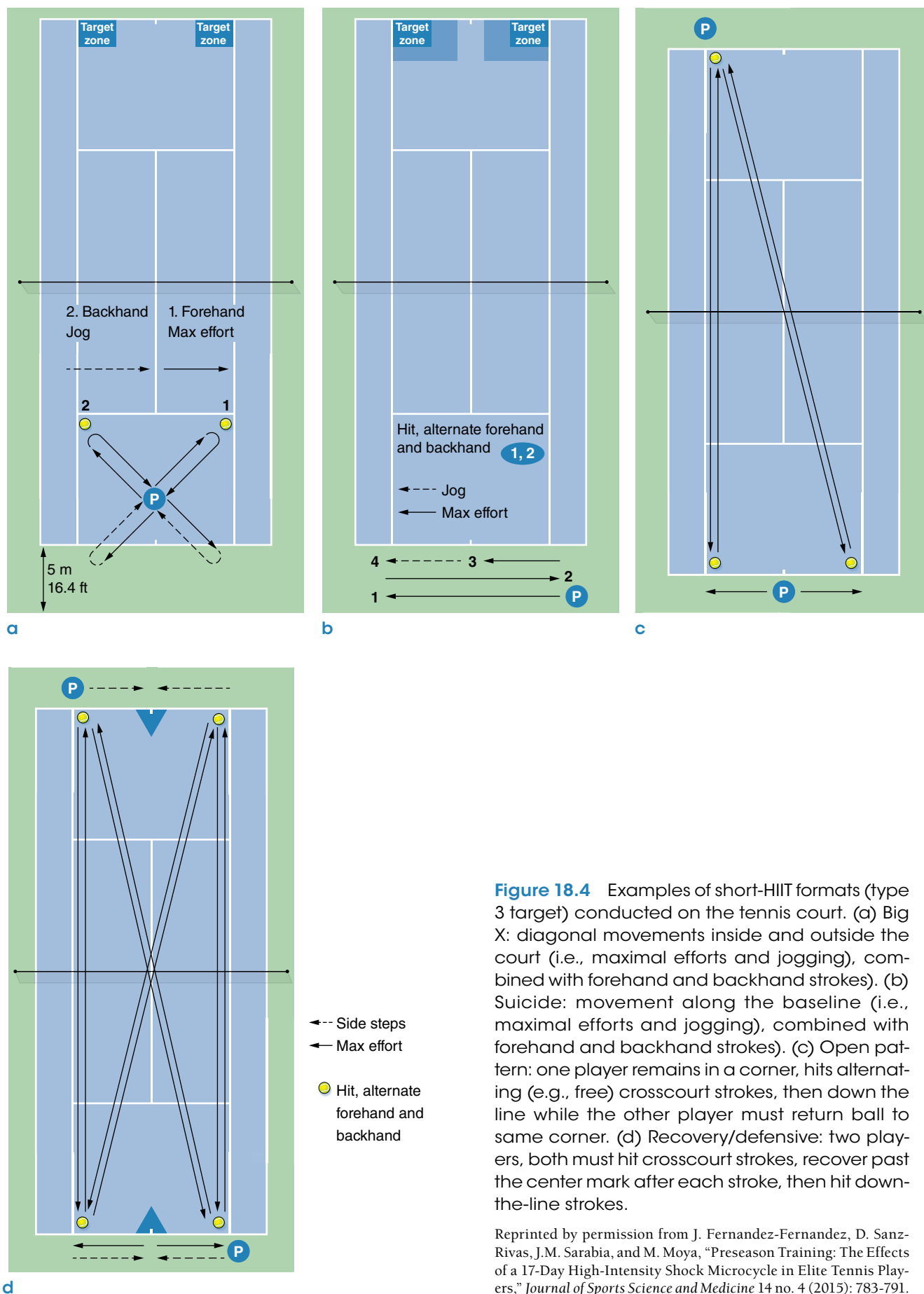


Figure 18.4 Examples of short-HIIT formats (type 3 target) conducted on the tennis court. (a) Big X: diagonal movements inside and outside the court (i.e., maximal efforts and jogging), combined with forehand and backhand strokes). (b) Suicide: movement along the baseline (i.e., maximal efforts and jogging), combined with forehand and backhand strokes). (c) Open pattern: one player remains in a corner, hits alternating (e.g., free) crosscourt strokes, then down the line while the other player must return ball to same corner. (d) Recovery/defensive: two players, both must hit crosscourt strokes, recover past the center mark after each stroke, then hit down-the-line strokes.

Reprinted by permission from J. Fernandez-Fernandez, D. Sanz-Rivas, J.M. Sarabia, and M. Moya, "Preseason Training: The Effects of a 17-Day High-Intensity Shock Microcycle in Elite Tennis Players," *Journal of Sports Science and Medicine* 14 no. 4 (2015): 783-791.

performance component, as most tennis movements occur within a 3 to 4 m radius, and rarely do players reach maximum speed (38). I also introduce short sprints with shuttles during the RST sequences, as from an injury-prevention perspective, these reduce stride length and subsequent hamstring strain risk (3). From my practical experience, a common mistake observed within training programs of tennis players around the world is using sprints greater than 20 m. With these, practitioners are often striving to obtain maximum sprinting speed. Instead, these often increase hamstring strain risk, when in reality, a tennis player rarely gets injured in this region of the body over the course of tennis matches (41).

The large neuromuscular and anaerobic responses of RST must be critically considered within a tennis player's programming puzzle (47). Such sessions should be conducted in isolation to other related content, as RST may affect the quality of subsequent sequences or sessions, as well as increase injury risk. We recently showed in top junior tennis players (16) that a combined RST and plyometrics training program, performed twice a week on top of regular tennis training sessions, led to substantial ($ES=0.56$ to 1.12) improvements in sprint (-2%), jump ($+2.2\%$), and repeated sprint performance (-1.4%). Thus, coaches can implement regular training sessions during competitive periods using a combined RST interspersed with at least 45 s of passive recovery. A sample RST program might include 3 or 4 sets of multidirectional short sprints (forward and side-to-side shuttles with one or two changes of direction, 5 or 6×15 to 20 m), interspersed with short (25 s) active recovery periods, and plyometric training (3 or 4 sets of 4 to 6 bodyweight exercises, bilateral and unilateral countermovement jumps [CMJ] to a 20 cm box [multidirectional], multilateral hops [hurdles], plyometric jumps [hurdles], step multilateral calf jumps, agility drills [ladders], and resisted standing start sprints [multidirectional]). Figure 18.5 shows some on-court drills I use to integrate as RST sequences conducted before the tennis-specific session.

Sprint Interval Training

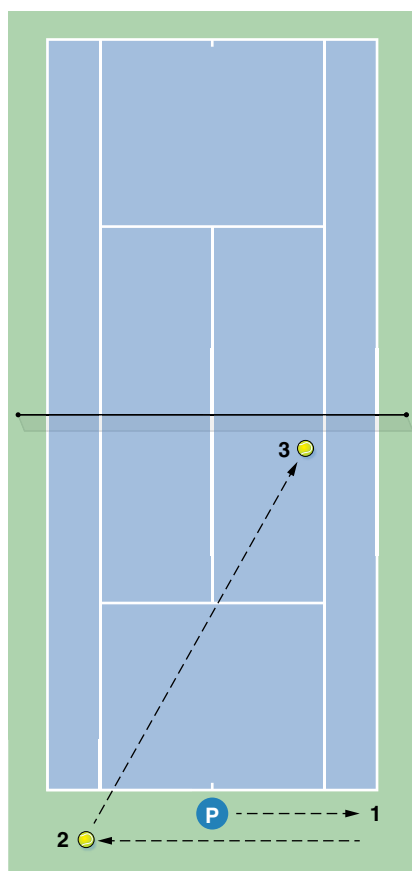
Generally the anaerobic glycolytic energy system is likely to be less involved during competitive tennis match play compared to popular belief. Thus, most training drills should be designed

accordingly, using short periods of maximal-intensity exercise with adequate rest between efforts to allow athletes to reproduce maximal or near-maximal performance in subsequent bouts. However, and especially at the elite level, increases in circulating [La] levels (up to 10 mmol/L) can occur under actual match play conditions during long and intense rallies, suggesting increased participation of glycolytic processes to energy supply (35, 36). Thus, SIT drills (efforts lasting 30 to 45 s; type 5 target) (23, 43) should be regularly administered to prepare a player to perform high-intensity exercise across these relevant durations. Specific on-court movements (preferably without the racket) are preferred to ensure attainment of the desired (maximal) intensities so that local muscle responses (proxy adaptations) can be fully transferred to actual match play. Due to the high anaerobic contribution and large neuromuscular fatigue associated with SIT, care of programming sessions during the week should follow a similar logic as that discussed for RST.

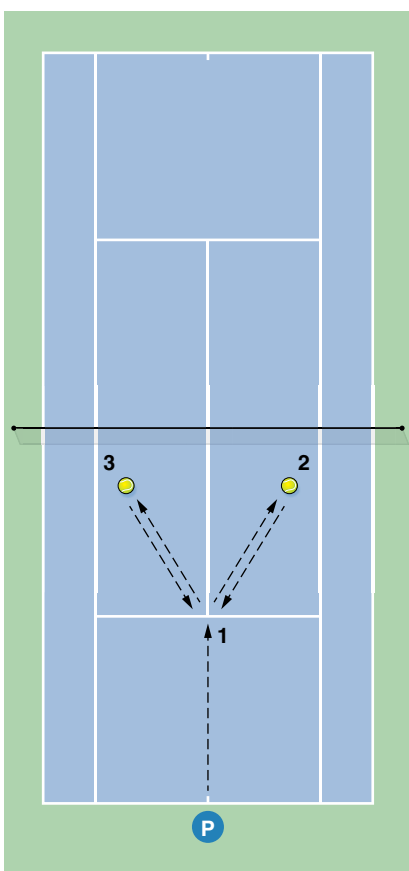
Game-Based HIIT

Considering that the maintenance of technical skill is a determining factor for success in tennis, and with training time always at a premium, I believe an integrated approach to conditioning and skill-based work is the most efficient training method for elite tennis players. This approach often results in the programming of game-based drills that include both technical (skills) and tactical (playing against an opponent) components as part of sport-specific conditioning (18). To facilitate this, I generally implement game-based HIIT (HIIT type 2, 3, and 4 targets, as detailed in figure 1.5 (5)). While run-based long intervals are generally associated with substantial lactate accumulation (type 3 or 4), the specific modality of tennis-specific HIIT with long intervals that includes a more intermittent neuromuscular activity (short rests between motor actions versus continuous and repeated contractions when running) likely allows for a lower anaerobic energy contribution. For that reason, some tennis-specific HIIT with long intervals can also be considered type 2 responses.

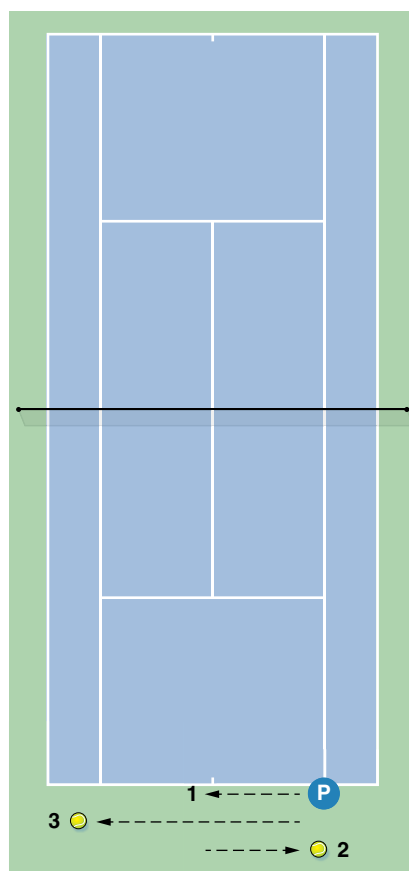
Since exercise intensity for game-based HIIT cannot be controlled precisely using percentage of V_{IFT} , opponent players are normally instructed to perform a variable number of 1 to 2 min drill-based HIIT bouts, depending on their level (e.g., pro players



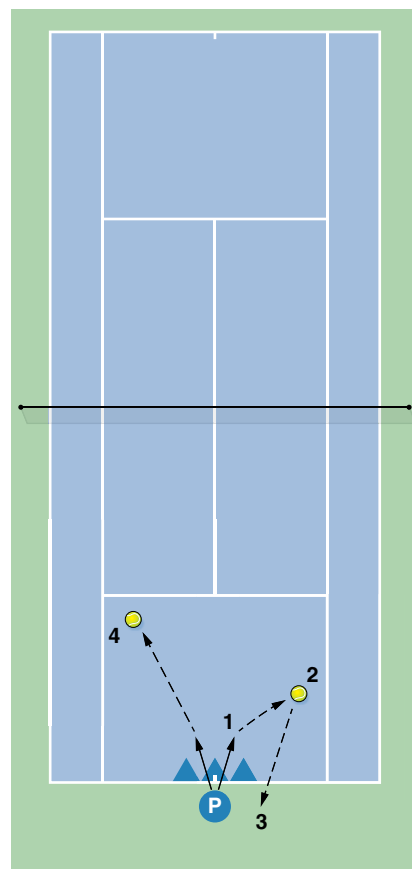
a



b



c



d

Figure 18.5 Examples of a repeated-sprint sequence format (type 4 target), in which players sprint in different directions (see number sequences), return to the center, and perform different strokes. Exercises last around 10 s, with 20 to 25 s rest, and several players can perform the exercise at the same time. P: player; 1-4: sprints.

are able to perform 8) at intensities between 90% and 95% of HR_{max} interspersed with 45 s to 90 s of passive recovery. Intensity and duration of the work bouts and recovery periods employed during these drills are selected based on prior experience, pilot studies, and previously published work (8, 31). Normally speeds are chosen individually using a trial-and-error approach to allow players to sustain the highest possible speed across all work bouts.

To ensure that all players perform at the desired intensity, we currently must rely on both HR monitoring and a player's motivation to ensure he commits to the required (likely maximum) intensity during the drills. For that reason, encouragement and scoring aims during such drills are key factors to apply, especially in professional players. It is also important to take into consideration that these drills are usually conducted with a maximum of three to four players on court, so when the number of players is higher, we need to combine training sessions (i.e., including HIIT short-interval runs at the same time using the lateral sides of the tennis court). Moreover, since players of varying fitness can train together, the relative work of the fitter players can wind up being relatively lower than that of the less fit, as happens in other intermittent sports. Therefore, the inclusion of extra tasks for fitter players can be implemented as follows:

- Before the beginning of the drill, the fittest player performs several short HIIT runs (e.g., 90% to 95% V_{IFT}) or a few short maximum sprints on the lateral side of the court.
- Before the drill commences, the fitter player performs a series of plyometric exercises.

An example of a typical on-court drill used in my training programs is shown in figure 18.6. Here players are required to perform forehand and backhand strokes in different positions for a work period of 1 to 2 min, with an experienced coach feeding balls (e.g., racket feed) outside the court. There are two different roles for both players; one is always playing inside the “defensive box,” trying not to make easy mistakes and always playing in a “defensive style,” while the other player must “attack” (offensive style), playing outside the defensive box. The coach feeds balls to the players when they make mistakes. Both players are required to move as fast as possible while playing their roles (i.e., defensive or offensive) until

the coach decides to change it up (i.e., normally after 45 s to 1 min). Intensity is monitored via HR monitoring, with targets set at 90% to 95% of HR_{max} (19). The number of balls played is counted as an indicator of work completed during each set. For scoring purposes, target areas can be implemented on the opposite side of the offensive player's court, and scoring points are assigned to each area. Thus, mean scores and percentage decrements for stroke accuracy can also be calculated for the drills. Training volumes rarely exceed a range of 5 to 8 repetitions, as reductions in accuracy are usually observed after the fifth set. Therefore, if the player requires more training volume, we can prescribe tennis-based HIIT with shorter exercises or run-based HIIT, using the V_{IFT} to prescribe intensity. However, in my experience, and also based on published work (17), these volumes appear optimal for obtaining improvements in markers of aerobic performance (e.g., $\dot{V}O_{2max}$, V_{IFT}) (5).

Longitudinal Training Effects of HIIT

Table 18.1 summarizes several studies recently conducted in nationally and internationally ranked male (junior and adult) tennis players (15-18). Overall, results showed that inclusion of HIIT programs during the preseason or in-season (2 or 3 times per wk, over 3 (shock microcycle) to 8 wk) resulted in small-to-large changes in physiological parameters (i.e., $\dot{V}O_{2peak}$) and performance-related variables (i.e., hit-and-turn tennis test; V_{IFT} ; sprint (5-20 m); CMJ; 5-0-5 agility test). Of course, it is difficult to compare the respective effects of the different interventions due to differences in training duration, baseline performance, and player characteristics. The research examining effects on different HIIT interventions in tennis remains scarce, and further studies comparing HIIT programs in players matched for baseline physical performance are required. Currently, we can presume that supplementing tennis with different HIIT sequences is likely to improve several performance-related parameters, including $\dot{V}O_{2max}$, V_{IFT} , performance in tennis-specific tests such as the hit-and-turn tennis test, or RSA. However, the relationships between these test-specific improvements and tennis performance (e.g., winning more matches or having a better ranking) is difficult to estimate, as tennis performance is highly complex.

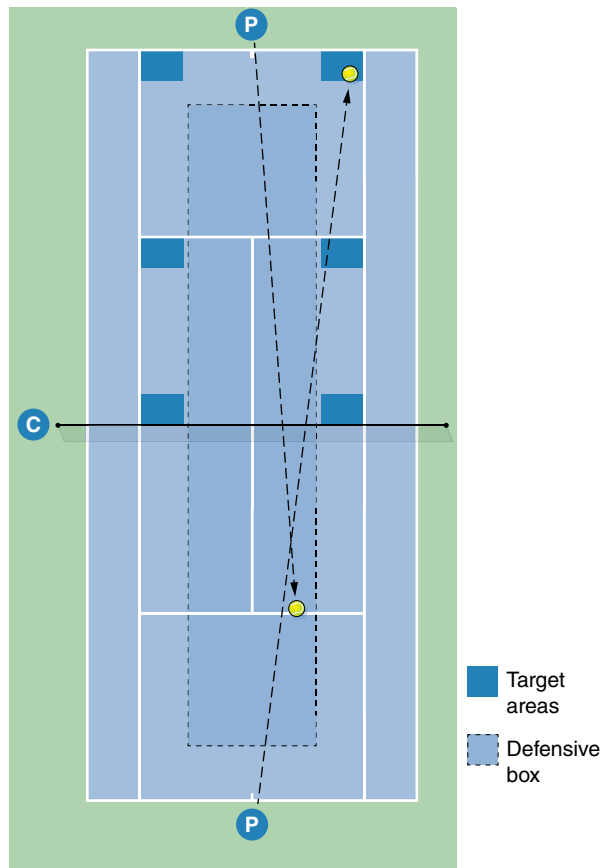


Figure 18.6 Example of game-based HIIT with long intervals (1 to 2 min work, 45 s to 1 min 30 s rest) (type 2 target). P: player; C: coach.

Load Surveillance and Monitoring Tools

Despite the very recent developments of monitoring tools (e.g., GPS, tracking systems), the use of wearables and tracking technologies is still very limited in tennis and is mainly reserved for research purposes (39). In my opinion, there is little relevant research concerning training load monitoring in tennis available that is transferable into the real-life context of the daily training environment. For the last 10 years, and despite their limitations, I have been using HR monitoring in a systematic way, together with the use of session RPE, to calculate the training load associated with tennis training (39). To be more specific with training load effects, I use a local RPE scale for both upper and lower body, the OMNI Scale of perceived exertion for

young players, the total quality recovery (TQR) (6-20 Likert scale), and/or the perceived muscle and joint soreness (1-10 Likert scale) (11, 17) in order to derive more detailed information concerning a player's fatigue state.

Training Status Surveillance and Monitoring Tools

As mentioned, my basic monitoring tools in tennis involve noninvasive and low-cost measurements, including questionnaires (revised Competitive Sport Anxiety Inventory (20)), jumps (neuromuscular status) using apps (e.g., MyJump (1)), and training HR (11). Lately, we've been using a general musculoskeletal assessment protocol every 1 to 2 wk and after several matches or tournaments. This involves static measurements and clinical examinations of joint integrity (e.g., hip and shoulder), ROM tests (e.g., passive assessment), together with tests of muscular strength patterns, also in the shoulders and hips (24, 37). Additionally, we often use the HR and RPE responses to HIIT sequences as markers of adaptations (generally, the lower the better the player's fitness) (6).

Strategies for Structuring the Training Program Using HIIT

In order to structure a training program in tennis, several factors must be taken into account. These include numerous controllable and uncontrollable factors, which obviously affect the progression and periodization of a training program involving HIIT.

Controllable and Uncontrollable Factors

Tennis is a complex sport, and this makes the process of planning, programming, and periodization highly challenging for the practitioner. The main difficulty revolves around a number of uncontrollable factors. The first uncontrollable factor is the player's competitive schedule. It is not only about the number of tournaments that are played each year, or the locations of tournaments, but more the number of matches played in each tournament. Due to the nature of the sport, this is largely unpredictable, making the sport even more complex from a programming standpoint than the team sport situation. More specifically, one of the main problems we have

Table 18.1 Training Interventions Using Various HIIT Formats in Tennis Players

Reference	Participants	Training intervention	Frequency	Results
20	31 nationally ranked male players	HIIT (long intervals; type 2 target) 3 × (3 × 90 s) level max* (90% to 95% HR _{max}), 3 min active recovery (70% of HR _{max}) between repetitions, 8 min of an on-court tennis game (i.e., 2:1 game) between sets versus RST (type 4 target) 3 × 10 to 20 m shuttle sprints, 20 s passive recovery between repetitions, 8 min of an on-court tennis game (i.e., 2:1 game) between sets	3 d/wk for 6 wk	For HIIT, $\dot{V}O_{2peak}$ +4.9% (ES=0.8), hit-and-turn test +28.9% (ES=1.6; significantly higher than RST), no changes in RSA test. For RST, $\dot{V}O_{2peak}$ +6% (ES=0.7), hit-and-turn test +14.5% (ES=1.2), RSA test mean time −3.8%.
14	16 internationally ranked competitive male junior players	Combined RST (type 5 target) 3 or 4 × 5 or 6 of 15 to 20 m sprints with 1 or 2 COD, 25 s active recovery plus 1 min 30 s rest with ExpS (3 or 4 sets of 4 to 6 exercises, 12 to 15 repetitions)	2 d/wk for 8 wk	Sprint (20 to 30 m) −2% to 3%, CMJ +2.2%, and RSA −1.4% were significantly improved (ES ranging from 0.56 to 1.12). No changes in V _{IFT} .
17	12 ATP-ranked male tennis players	HIIT (types 2, 3, and 4 target) shock microcycle combining on-court long intervals (7 × 2 min on/1 min 30 s off; 90% to 95% HR _{max}), short-interval runs (3 or 4 × 10 to 15 repetitions of 30 s on/30 s off; 15 s on/15 s off; 90% to 95% V _{IFT}), and on-court short and long intervals (30 s on/30 s off; 60 s on/60 s off; 90% to 95% HR _{max})	13 sessions in 17 d	V _{IFT} +6.5% (ES=1.9), RSA mean −0.5% (ES=0.4)
18	20 top-50 U15 nationally ranked tennis players	Mixed HIIT (type 3 and 4 targets), short intervals (15 s on/15 s off; 90% to 95% V _{IFT}) and on-court training of short and long intervals (30 s on/30 s off; 60 s on/60 s off; 90% to 95% HR _{max}) versus on-court training only (type 3 and 4 targets), short and long intervals (30 s on/30 s off; 60 s on/60 s off; 90% to 95% HR _{max})	2 d/wk for 8 wk	For mixed HIIT, $\dot{V}O_{2peak}$ +4.2% (ES=1.05), V _{IFT} +6.3% (ES=0.9), 5-0-5 test −2.6% (ES=0.8). For on-court only, $\dot{V}O_{2peak}$ +2.4% (ES=0.5), V _{IFT} +2.2% (ES=0.4).

* Levelmax: maximum level reached in the hit-and-turn tennis test (22); RSA: repeated-sprint ability; $\dot{V}O_{2peak}$: peak oxygen uptake; ES: effect size; COD: change of direction; ExpS: explosive strength; CMJ: countermovement jump; V_{IFT}: velocity of the 30-15 intermittent fitness test; U15: under 15.

in conducting periodized programs in tennis is the lack of consistency in tournament play, as the coach–player team will constantly adjust the tournament schedule depending on results. For example, if a player loses too many first-round matches in a row, the typical reaction is to play more tournaments instead of taking a break and readjusting competition loads, conducting a proper reconditioning program, and so forth.

The most controllable situation in tennis may be the preseason, when the strength and conditioning coach is typically primarily responsible for the training program, together with the coach. However, due

to the ever-increasing demands imposed on players, less and less training time tends to be devoted to preparation, with preseasons normally reduced to only 5 to 7 wk. At the highest levels of the game (i.e., ATP/WTa players or even top junior players), this preseason may be reduced further with the increase in exhibition events or team events and tournaments. Regardless, during the preseason, most tennis players prioritize fitness training during the first couple of weeks, while the maintenance of technical and tactical skills is also important. Here, all HIIT types detailed earlier fit well into the training program, alongside clear communication with the coach.

Training content is not only about HIIT weapons. Integrating strength and power training, speed and agility training, neuromuscular (e.g., injury prevention) training, as well as technical and tactical sessions is also needed. Thus, the HIIT sessions should be limited in terms of interfering with subsequent training sessions. Therefore, it is important to take into account the residual neuromuscular fatigue post-HIIT through the use of monitoring tools (see previous sections and chapters 8 and 9) (35).

For me, another uncontrollable factor is the in-season training schedule, as the match schedule will more or less dictate the training program. This creates many possible scenarios to take into account. For example, you can have a player playing qualifying rounds (2 or 3 matches, depending on the tournament), and if successful, he will play in the main draw (a week schedule, with 6 or 7 matches). When the player finishes in the tournament, the time until the next tournament dictates whether or not we have room for HIIT. Nevertheless, I tend to program 1 or 2 HIIT sessions per week when a full week between tournaments arises in the schedule.

Program Structure and Progression

As previously mentioned, and similar to other intermittent sports, very little periodization occurs in elite tennis, except during the preseason (see tables 18.2 and 18.3). The weeks available to conduct

a recovery or reconditioning program dictate the periodization level of the design. In doing so, I follow some basic steps to create a tennis conditioning training program.

First, after the number of tournaments to play has been decided, select those tournaments at which the athlete needs to be in peak form and highlight them on the calendar. Players can play tournaments every week, and it is critical to schedule the tournaments appropriately to minimize the chance of injury and potential burnout (10). There are some recommendations for training and competition volume for junior tennis players that suggest that players at the junior stage of development should consider playing less than two tournaments per month, and less than 18 tournaments annually in the U18 category, in order to reduce injury risk (32). From here, I will recommend planned structures for the season (i.e., mesocycles), including specific goals. Thus, I normally suggest that coaches and players have 5 to 7 wk of preseason, and introduce 2 or 3 blocks of training (e.g., mesocycles) of 2 to 3 wk each in order to somewhat “periodize” their training. In-season, when a complete week is available to train, I attempt to keep similar weekly training loads throughout the months and focus on successive tournament preparation-match-recovery cycles. A typical example of an in-season weekly training program is shown in table 18.4.

Table 18.2 Sample Preseason Program (First Week(s) of Preseason, Typically 1 to 2 wk) of an Elite Tennis Player

Microcycle (physical emphasis)		Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Aerobic ($\dot{V}O_2$) development	AM	Injury prevention	Strength training	Injury prevention	Strength training	Injury prevention	Injury prevention	Rest
		Tennis		Tennis		Tennis	Tennis	
	PM	Type 2 HIIT long (on-court) 5 or 6 × 2 min at 90% to 95% HR_{max} , 1 min 30 s rest	Tennis	Tennis	Tennis	Type 3 HIIT short (on-court) 15 min of short and long intervals (30 s on/30 s off, 60 s on/60 s off, at 90% to 95% HR_{max})	Neuromuscular training (e.g., plyometrics, speed/agility training)	
				Type 3 HIIT short (runs) 2 × 16 × 15 s on/15 s off at 90% V_{IFT} , 3 min rest			Tennis	

Table 18.3 Sample Preseason Program (Last Week(s) of Preseason, Typically 1 to 2 wk) of an Elite Tennis Player

Microcycle (physical emphasis)		Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Aerobic ($\dot{V}O_2$) development	AM	Injury prevention	Strength training	Injury prevention	Tennis	Injury prevention	Neuromuscular training (e.g., plyo- metrics, speed/ agility training)	Rest
		Tennis		Tennis		Tennis		
	PM	Tennis	Neuromuscular training (e.g., plyo- metrics, speed/ agility training)	Type 4 HIIT RST 3 × (5 or 6 × 15 to 20 m with 2 or 3 shuttles), 25 s rest between repetitions, 4 min rest between sets	Strength training	Type 3 HIIT (on-court) 20 min of short and long intervals (30 s on/30 s off, 60 s on/60 s off, at 90% to 95% HR _{max})	Strength training	
		Type 3 HIIT (runs) 2 × 22 × 15 s on/15 s off at 95% V _{IFT} , 3 min rest						

Incorporation of HIIT

During the preseason and based on testing results (e.g., 30-15 IFT (4), which I normally conduct over 28 m (30)), I include the long- and short-interval HIIT programs as part of the tennis training. Therefore, conditioning goals are integrated into the tennis practice in order to minimize the possible fatigue effects on subsequent training. If this is not possible, running-based HIIT can be conducted after tennis sessions (table 12.2).

For in-season routines, when a complete week of training is available, the HIIT sessions are incor-

porated into the tennis training, including more neuromuscular-demanding HIIT such as RST. The main goal of these sequences is to maintain the fitness (cardiorespiratory and neuromuscular) levels.

Sample Training Programs

Tables 18.2 and 18.3 provide examples of HIIT programming during preseason training, while table 18.4 provides an example of in-season programming in tennis.

Table 18.4 Sample Maintenance Schedule (In-season, Remainder of the Season, Through the Competitive Period, When a Full Week Is Available) of an Elite Tennis Player

Microcycle (physical emphasis)		Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Competitive (in season)	AM	Injury prevention	Strength training	Injury prevention	Injury prevention	Injury prevention	Neuromuscular training (e.g., plyometrics, speed/agility training)	Rest
			Tennis		Tennis		Tennis	
	PM	Type 4 HIIT RST 3×5 or 6×15 to 20 m with 2 or 3 shuttles, 25 s rest between repetitions, 4 min rest between sets	Injury prevention and recovery	Neuromuscular training (e.g., plyometrics, speed/agility training)	Type 4 HIIT RST 3×5 or 6×15 to 20 m with 2 or 3 shuttles, 25 s rest between repetitions, 4 min rest between sets or Type 5 5 or 6×25 to 35 s all-out sequences, 5 min rest (low- intensity play)	Tennis	Strength training	
					Tennis*		Tennis	

* When the HIIT sequence conducted is type 5, the following tennis training session is normally canceled.

CASE STUDY

I've been privileged to work as a strength-and-conditioning coach, as well as a sports science consultant, for several top-100 ATP/WTa players and their coaches for the last few years. There is an interesting story about how I was able to integrate fitness into the tennis practice through the use of HIIT sequences. A few years ago, I began working with a "skeptical" coach, who for many years, defended his belief that professional tennis players needed to run and accumulate many miles of threshold-based (i.e., 80% HR_{max}) endurance runs. His view was supported by tradition and acceptance in tennis that, "Running is not only important for performance, but also for the head." Like many sports, tennis is filled with many longstanding traditions in the realm of physical training, and it can be difficult for practitioners to change these traditional preparation practices, despite what science or common sense may tell us otherwise. Related to this particular story is an old belief that, "tennis training is just tennis," and for some coaches this means no additional physical exertion during the tennis training. As you might imagine, this particular player encountered many situations in which he was doubling his training load (e.g., on- and off-court). The coach, however, continued to defend his player's need to run and accumulate the traditional endurance-based runs.

My intervention with this particular player was aimed at showing both the coach and the player that training volume could be substantially lowered with a new focus on improving injury prevention. We started by analyzing the short-term variations in the player's cardiorespiratory fitness after 6 wk of tournaments (e.g., average of 3 or 4 matches per week, plus training). The player was not allowed to conduct any endurance sessions during this time, apart from his specific tennis training (which, for the coach, was not an aerobic stimulus). We found that his $\dot{V}O_{2max}$ levels didn't change much (from 58 to 57 $mL \cdot kg^{-1} \cdot min^{-1}$). What I did instead was implement integrated HIIT sessions (as explained) into his regular tennis schedule for the next preseason. The player was not allowed to conduct his traditional supplemental aerobic sessions during the 7 wk period of the preseason, which can be psychologically challenging for a player due to the traditional routine he is used to, potentially leading to loss of self-confidence from a perceived fitness level perspective (i.e., "I'm not working enough"). This situation can be a challenge for the strength-and-conditioning coach to field. After 7 wk and about a 30% to 40% reduction in training volume, the player was physically better than ever. $\dot{V}O_{2max}$ values increased slightly (58.2 to 60.3 $mL \cdot kg^{-1} \cdot min^{-1}$), V_{IFT} improved from 20 to 21.5 km/h, and he increased HIIT work from about 40 min to more than 50 min. Moreover, the player also increased target accuracy (by about 45%) in the drills I was monitoring.

Unfortunately, the coach remained skeptical, and we ended our collaboration due to our "different visions of tennis training." Nevertheless, I felt that my aims were achieved, with physical fitness and tennis-specific performance improved. The player remained with his coach, although he recognized that his fitness level was better than ever. The player tells me that he continues running for about an hour (70% to 80% HR_{max}) three times per week.

The conclusions we can draw from this story may be that, sometimes, and especially when you're working in a high-performance sport, tradition and beliefs often trump evidence or science-based logic. However, we can continue to fight against these beliefs using our evidence-based work, with change sometimes just requiring more time and a changing of the guard. While admittedly this is difficult and challenging, it is possible to introduce coaches and players into a more-controlled training environment, using for example, monitoring tools (i.e., chapters 8 and 9). Moreover, when you're part of a strong team and the coach and player are on your side, it is possible to plan and control all the physical training loads (and also recovery schedules) in the preparation of the player. Working together and assisting the coach (e.g., using tennis-based HIIT) alongside monitoring the training loads (e.g., HR, RPE) is key.

This page intentionally left blank